Bank Customer Churn Analysis

Pivot Table Analysis Report

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Tool Used: Microsoft Excel
Dataset: Bank Customer Churn Dataset (Maven Analytics)

Table of Contents

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1. Overall Customer Churn Analysis

Business Question

What is the overall customer churn rate?

Objective

Determine the total number of customers who have churned and calculate the overall churn rate.

Pivot Table

Total Customers	
Exited	Total
0	7963

1	2037
Grand Total	10000

Key Findings

- Total Customers:
- Churned Customers:
- Churn Rate:

Business Insight

Explain what the numbers mean for the business.

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2. Gender-Based Churn Analysis

Business Question

Does customer churn differ by gender?

Objective

Compare churn rates between male and female customers.

Pivot Table

Gender-Based Churn Analysis	Gender		
Cities	Female	Male	Grand Total
France	2261	2753	5014
Germany	1193	1316	2509
Spain	1089	1388	2477
Grand Total	4543	5457	10000

Key Findings

- Female Churn Rate:
- Male Churn Rate:

Business Insight

State which gender shows higher churn and whether the difference is significant.

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3. Geographic Churn Analysis

Business Question

Which geographic region experiences the highest customer churn?

Objective

Compare churn rates across countries.

Pivot Table

Count of Exited	
Geography	Total
France	5014
Germany	2509
Spain	2477
Grand Total	10000

Key Findings

- Highest Churn Region:
- Lowest Churn Region:

Business Insight

Explain possible business implications.

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4. Age Group Churn Analysis

Business Question

How does customer age influence customer churn?

Analysis Objective

- Segment customers into different age groups.
- Compare churn rates across age categories.
- Identify the age group most likely to leave the bank.

Pivot Table

Age Group Churn Analysis	
Age Group	Total
Adult	557
Senior	1440
Younger	40
Grand Total	2037

Key Findings

- Customers were grouped into age categories for comparison.
- **Senior customers exhibited the highest churn rate** among all age groups.
- Younger and adult customers showed comparatively lower churn rates.

Business Insight

Customer age has a significant influence on churn behavior. Senior customers appear to be the most vulnerable segment, suggesting that the bank should better understand their financial needs and engagement patterns.

Business Recommendation

- Develop personalized banking products for senior customers.
- Conduct satisfaction surveys to understand reasons for churn.
- Offer loyalty rewards and dedicated customer support for older customers.

Conclusion

Age is an important factor influencing customer churn, with senior customers representing the highest-risk age segment.

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5. Credit Card Ownership Analysis

Business Question

Does credit card ownership affect customer retention?

Analysis Objective

- Compare churn rates between customers with and without credit cards.
- Determine whether credit card ownership contributes to customer loyalty.

Pivot Table

	Data		
HasCrCard	Total Customers	Churned Customers	Churn Rate
No Credit Card	7055	1424	20.18%
Has Credit Card	2945	613	20.81%
Grand Total	10000	2037	20.37%

Key Findings

- Customers **without a credit card** had a churn rate of **20.8%**.
- Customers **with a credit card** had a churn rate of **20.2%**.

- The difference between the two groups is minimal.

Business Insight

Credit card ownership does not significantly impact customer churn. Other factors such as customer engagement, product usage, and satisfaction are likely to have a stronger influence on retention.

Business Recommendation

- Focus retention efforts on behavioral factors rather than credit card ownership.
- Enhance credit card value through rewards and personalized benefits.

Conclusion

Credit card ownership alone is not a reliable predictor of customer churn.

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6. Customer Engagement Analysis

Business Question

Are inactive customers more likely to churn?

Analysis Objective

- Compare churn rates between active and inactive customers.
- Assess the relationship between customer engagement and churn.

Pivot Table

Customer Engagement Analysis		
IsActiveMember	Total	Churn_Rate
Not Active	1302	64%
Active	735	36%
Grand Total	2037	

Key Findings

- Active Customers: **5,151**
- Inactive Customers: **4,849**
- Active Customer Churn Rate: **14.3%**
- Inactive Customer Churn Rate: **26.9%**

Business Insight

Inactive customers are almost twice as likely to leave the bank compared to active customers. This indicates that customer engagement is one of the strongest predictors of churn.

Business Recommendation

- Launch engagement campaigns targeting inactive customers.
- Encourage regular account activity through rewards and personalized offers.
- Monitor inactivity as an early warning sign of churn.

Conclusion

Improving customer engagement can significantly reduce customer churn.

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7. Balance-Based Churn Analysis

Business Question

Which account balance segment has the highest customer churn rate?

Analysis Objective

- Categorize customers into balance segments.
- Compare churn rates across different balance categories.
- Identify high-value customers at risk of leaving.

Pivot Table

Balance-Based Churn Analysis	
Balance_Classes	Total
100K+	1211
1-50K	26
50K-100K	300
Zero Balance	500
Grand Total	2037

Key Findings

- Customers were segmented based on account balance.
- Churn behavior varies across balance categories.
- High-balance customer churn should be closely monitored due to its potential financial impact.

Business Insight

Account balance helps identify valuable customer segments. Losing customers with higher balances can have a greater impact on the bank's revenue than losing low-balance customers.

Business Recommendation

- Prioritize retention efforts for high-balance customers.

- Offer premium banking services and relationship management to high-value clients.

Conclusion

Balance segmentation enables the bank to identify financially valuable customers who may require targeted retention strategies.

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8. Product Usage Analysis

Business Question

Does owning more banking products improve customer retention?

Analysis Objective

- Compare churn rates across customers with different numbers of banking products.
- Evaluate the relationship between product ownership and customer loyalty.

Pivot Table

Product Usage Analysis	
NumOfProducts	Total
1	1409
2	348
3	220
4	60
Grand Total	2037

Key Findings

- Customer churn varies according to the number of products owned.
- Product ownership influences customer retention behavior.

Business Insight

Customers who utilize multiple banking products are generally more engaged with the bank, while customers with fewer products may be more likely to switch to competitors.

Business Recommendation

- Increase cross-selling opportunities.
- Encourage customers to adopt additional banking products through bundled offers.

Conclusion

Product ownership is an important indicator of customer engagement and retention.

9. Credit Score Analysis

Business Question

How does credit score influence customer churn?

Analysis Objective

- Group customers by credit score.
- Compare churn rates across different credit score categories.

Pivot Table

Credit Score Analysis	
Category	Total
Excellent	128
Fair	753
Good	496
Poor	660
Grand Total	2037

Key Findings

- Customers were categorized into Poor, Fair, Good, and Excellent credit score groups.
- Churn patterns differ among credit score categories.

Business Insight

Credit score provides useful information about customer financial profiles. Customers at the extremes of the credit score range may exhibit different churn behaviors, helping the bank identify segments that require different retention approaches.

Business Recommendation

- Develop tailored financial products for different credit score segments.
- Monitor high-risk credit score groups more closely.

Conclusion

Credit score contributes to understanding customer behavior and supports targeted retention strategies.

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10. Salary-Based Churn Analysis

Business Question

Does customer salary influence churn behavior?

Analysis Objective

- Segment customers by salary range.
- Compare churn rates across income groups.

Pivot Table

Salary-Based Churn Analysis	
Salary Groups	Total
<50K	489
100K-150K	517
150K+	527
50K-100K	504
Grand Total	2037

Key Findings

- Customers were grouped into salary segments.
- Churn behavior was analyzed across different income levels.

Business Insight

Salary alone does not fully explain customer churn. It should be considered alongside engagement, product ownership, and other customer characteristics for a more complete understanding of churn risk.

Business Recommendation

- Design banking products that align with the needs of different income groups.
- Combine salary analysis with other customer attributes to improve targeting.

Conclusion

Salary is one of several factors influencing churn and should be analyzed together with behavioral metrics.

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11. High-Value Customer Risk Analysis

Business Question

How many high-value customers has the bank lost?

Analysis Objective

- Identify customers with high balances and high salaries.

- Evaluate churn among premium-value customers.

Pivot Table

High-Value Customer Risk Analysis	Salary Groups				
Balance_Classes	<50K	100K-150K	150K+	50K-100K	Grand Total
100K+	307	303	309	292	1211
1-50K	4	6	8	8	26
50K-100K	60	80	82	78	300
Zero Balance	118	128	128	126	500
Grand Total	489	517	527	504	2037

Key Findings

- High-value customers were identified using balance and salary criteria.
- Customer churn within this segment was evaluated to estimate business risk.

Business Insight

The loss of high-value customers has a greater financial impact than the loss of average customers. Monitoring this segment enables the bank to prioritize retention efforts where they have the highest business value.

Business Recommendation

- Assign relationship managers to high-value customers.
- Provide exclusive benefits and personalized financial solutions.

Conclusion

Retaining premium customers is essential for protecting revenue and strengthening long-term profitability.

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12. Geography & Gender Analysis

Business Question

Which geography and gender combination experiences the highest customer churn?

Analysis Objective

- Compare churn across geographic regions and gender.
- Identify high-risk customer segments.

Pivot Table

Geography and Gender Analysis	Gender		
Geography	Female	Male	Grand Total
France	460	350	810
Germany	448	366	814
Spain	231	182	413
Grand Total	1139	898	2037

Key Findings

- Customer churn varies across both geography and gender.
- Certain region-gender combinations show higher churn than others.

Business Insight

Combining demographic dimensions provides a deeper understanding of customer behavior and enables more targeted retention strategies.

Business Recommendation

- Develop region-specific marketing campaigns.
- Personalize communication based on customer demographics.

Conclusion

Geography and gender together provide valuable insights into customer churn patterns.

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13. Age & Product Combination Analysis

Business Question

Which age group and product ownership combination is most likely to churn?

Analysis Objective

- Compare churn across age groups and product ownership.
- Identify customer segments requiring targeted retention.

Pivot Table

Age and Product Combination Analysis	NumOfProducts				
Age Group	1	2	3	4	Grand Total
Adult	376	105	63	13	557
Senior	1008	232	153	47	1440
Younger	25	11	4		40
Grand Total	1409	348	220	60	2037

Key Findings

- Churn patterns differ across combinations of age and product ownership.
- Certain customer segments demonstrate higher churn risk.

Business Insight

Combining demographic and behavioral factors provides a more comprehensive understanding of customer churn than analyzing either factor independently.

Business Recommendation

- Offer personalized product bundles to high-risk age groups.
- Strengthen cross-selling initiatives for customers with limited product usage.

Conclusion

Age and product ownership together help identify customer segments requiring focused retention efforts.

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14. Customer Tenure Analysis

Business Question

Are new customers more likely to churn than long-term customers?

Analysis Objective

- Compare churn rates across different customer tenure groups.
- Evaluate customer retention throughout the customer lifecycle.

Pivot Table

Customer Tenure Analysis

	Data		
Tenure Classes	Count of CustomerId	Sum of Exited	Churn Rate
Long-Term Customer	1474	314	21%
Loyal Customer	3020	570	19%
Mid-Term Customer	3010	625	21%
New Customer	2496	528	21%

Key Findings

- Churn rates remain relatively consistent across tenure groups.
- No tenure segment exhibited significantly higher churn than others.

Business Insight

Customer tenure alone is not a strong predictor of churn. Retention strategies should focus more on customer engagement and satisfaction than simply the length of the customer relationship.

Business Recommendation

- Continue engaging customers throughout their lifecycle.
- Combine tenure analysis with behavioral metrics for more accurate churn prediction.

Conclusion

Customer tenure has a limited impact on churn compared to other customer characteristics.

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15. Customer Churn Persona Development

Business Question

What are the common characteristics of customers most likely to churn?

Analysis Objective

- Combine insights from all analyses to develop a customer churn persona.
- Identify the characteristics of high-risk customers.

Key Findings

- Customers most likely to churn are typically:
    - Senior age group
    - Inactive members

    - Associated with other high-risk characteristics identified through segmentation analysis

Business Insight

Creating a churn persona helps the bank move from reactive reporting to proactive customer retention. By understanding common characteristics, the bank can identify at-risk customers earlier and implement targeted interventions.

Overall Business Recommendations

- Improve engagement among inactive customers.
- Develop targeted retention campaigns for senior customers.
- Monitor high-risk customer segments regularly.
- Increase customer loyalty through personalized banking services.
- Use data-driven dashboards to support strategic decision-making.

Conclusion

This analysis demonstrates how Microsoft Excel can be used to transform raw banking data into meaningful business insights through Pivot Tables, Pivot Charts, calculated metrics, and dashboard reporting. The findings provide valuable recommendations that can help reduce customer churn and improve customer retention.